LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 4. Operators and Expressions

Lecture: 32. Challenge-Surface area of Cuboid

Section 1: Lecture Summary

This lecture covers the concept of the total surface area of a cuboid, including understanding what a cuboid is, identifying its dimensions and faces, deriving the formula for total surface area, and applying the formula in Python programming. It begins by explaining the cuboid as a three-dimensional figure with length, breadth, and height and six faces. Then the lecture explains how to calculate the area of each face and sum them to find the total surface area. Finally, it demonstrates writing a Python program to compute the total surface area given the dimensions, starting with fixed values and suggesting an extension to accept user inputs.

Section 2: Key Concepts and Explanations

- What is a Cuboid?
  - A cuboid is a three-dimensional rectangular figure, similar to a box or brick.
  - It has three dimensions:
    - Length (l)
    - Breadth (b)
    - Height (h)
  - It has six rectangular faces: front, back, top, bottom, left, and right.

- Faces of a Cuboid
  - Front and Back Faces: Dimensions are length × height
  - Top and Bottom Faces: Dimensions are length × breadth

- Left and Right Faces: Dimensions are breadth × height

- Total Surface Area (TSA) of a Cuboid
  - Defined as the sum of the areas of all six faces.
  - Area of each pair of opposite faces:
    - Front & back: \(2 \times l \times h\)
    - Top & bottom: \(2 \times l \times b\)
    - Left & right: \(2 \times b \times h\)
  - The formula:

\[
\text{TSA} = 2 \times (l \times h + l \times b + b \times h)
\]

- Applying the Formula in Python
  - Variables to store length, breadth, and height
  - Calculate total surface area using the formula
  - Print the result
  - Optionally, accept user input to calculate TSA for any cuboid dimensions

Section 3: Example Code and Use Cases

Given dimensions of the cuboid

Calculate total surface area using the formula

Print the total surface area

```
length = 10
breadth = 5
```

```
height = 7

area = 2 * (length * height + length * breadth + breadth * height)

print('Area:', area)
```

Description: This code calculates the total surface area of a cuboid with length 10, breadth 5, and height 7, then prints the result.

Example: Taking input from the user to calculate TSA dynamically

Accept user input and convert to integers

Calculate total surface area

Display the result

```
length = int(input("Enter length: "))
breadth = int(input("Enter breadth: "))
height = int(input("Enter height: "))

area = 2 * (length * height + length * breadth + breadth * height)

print('Total Surface Area:', area)
```

Description: This code modifies the previous example to accept dimensions from the user, allowing calculation of the total surface area for any cuboid.

Section 4: Key Takeaways

- A cuboid is a 3D rectangular box with length, breadth, height, and six rectangular faces.

- Total surface area is the sum of the areas of all six faces.

- The TSA formula is \(2(lh + lb + bh)\).

- In Python, use variables for dimensions and apply the formula to compute TSA.

- Programs can be adapted to take dynamic input from users, improving flexibility.

- Practice by typing and running the code manually to reinforce understanding.